A 15.4-ppm/°C GaN-Based Voltage Reference With Process-Variation-Immunity and High PSR for Electric Vehicle Power Systems

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***Abstract*—The proposed gallium nitride (GaN)-based voltage reference (V_{REF}) generator has a low temperature coefficient (TC) of 15.4 ppm/°C, small V_{REF} deviation at different process corners (standard deviation of 0.22%), line sensitivity as low as 0.0023%/V, and high power supply rejection (PSR) of −187 and −114 dB at 100 Hz and 50 MHz, respectively. The proportional-to-absolute-temperature (PTAT) gate current for enhancement-mode GaN (eGaN) optimizes TC. Eliminating depletion-mode GaN (dGaN) gate leakage and using multiple stacked composite dGaNs can improve line regulation and PSR. All performance is achieved with a low power consumption of 10.9 μW.**

***Index Terms*—DC–DC converter, electric vehicle (EV), gallium nitride (GaN), GaN reference voltage, monolithic integration,**

I. Introduction

With the rapid growth of the electric vehicle (EV) industry, EV power systems are becoming increasingly crucial. As shown in Fig. 1(a), DC–DC converters play a key role in an EV and usually operate at high temperatures due to large load currents. State-of-the-art gallium nitride (GaN)-based voltage reference generators (VRGs) [1], [2], [3], [4], [5] are better suited for high-temperature operation (∼200 °C) in EV power systems compared to silicon components [6], [7] because of the wider bandgap of GaN devices. Additionally, DC–DC converters with GaN power switches operate at tens of megahertz [8], enabling compact designs and high-power density. However, the high-switching frequency of GaN power switches introduces significant dV/dt coupling noise into the supply voltage (V_{DD}), which may affect the stability of the VRG, as shown in Fig. 1(b). Consequently, VRGs in EV power systems must exhibit excellent power supply rejection (PSR) performance in the high-frequency domain to maintain stability when V_{DD} contains high-switching noise. Moreover, VRGs must have low-line regulation to operate effectively across a wide V_{DD} range of 4 to 48 V in EV power systems. Furthermore, VRGs must be resilient to process, voltage, and temperature (PVT) variations, which are significant in GaN manufacturing. For instance, the threshold voltage (V_{TH}) of enhancement-mode GaN (eGaN) and depletion-mode GaN (dGaN) at the slow-slow (SS) process corner can be two or more times larger than at the fast-fast (FF) corner.

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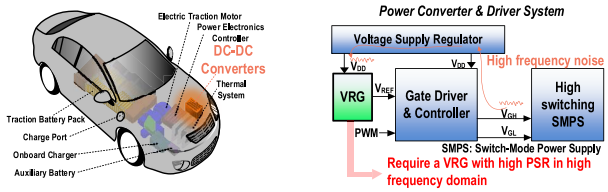

(a) (b)

Fig. 1. (a) Key components of an EV. (b) Applications and goals of proposed GaN-based VRG.

Based on the above depiction of the design challenges in EV applications, this letter proposes a monolithically integrated GaN-based VRG capable of suppressing PVT variations while maintaining low-power consumption [9]. Line regulation and PSR have also been improved to meet the requirements for high-power and high-voltage operation. This letter is organized as follows. Section II illustrates the problems in state-of-the-art GaN-based VRGs. In Section III, circuit implementation and optimization methods are discussed. Measurement results and the comparison table are presented in Section IV. Finally, the conclusions are made in Section V.

II. State-of-the-Art Defects in GaN Process Cause Deviations in Reference Voltage

State-of-the-art GaN-based VRGs face several critical challenges, including high-power consumption and significant V_{REF} variations, due to process and V_{DD} fluctuations. Fig. 2(a) compares power consumption and temperature coefficient (TC) with state of the arts. The single-branch VRGs in [2] and [5] achieve low power consumption of less than 21 μW, but their TC exceeds 110 ppm/°C. In contrast, Liao et al. [1] achieved a low TC of less than 36 ppm/°C, but its power consumption is twice that of [2] and [5] due to dual current paths. The 2-transistor GaN-based VRG shown in Fig. 3 also consumes substantial power [3]. At the FF corner, the absolute value of the dGaN threshold voltage ($|V_{THD}|$) is significantly higher than the eGaN threshold voltage (V_{THE}), causing V_{REF} to be clamped at V_{THE}. Consequently, the dGaN M_D operates with a large gate-to-source voltage (V_{GS}), leading to increased quiescent current (>5mA) and power consumption (>20mW).

Furthermore, Fig. 2(b) shows process variations result in [1] and [5] having lower TC but wider V_{REF} range, [2], [3] having converged V_{REF} but larger TC, and [4] having moderate TC but slightly wider V_{REF} distribution. In prior arts [1], [2], [5], V_{REF} is generated by

$$V_{REF} = M \times V_{THE} + N \times |V_{THD}| \tag{1}$$

where V_{THE} and $|V_{THD}|$ represent proportional-to-absolute-temperature (PTAT) and complementary-to-absolute-temperature (CTAT) voltages, respectively. For low TC, M is assumed to be 1 in

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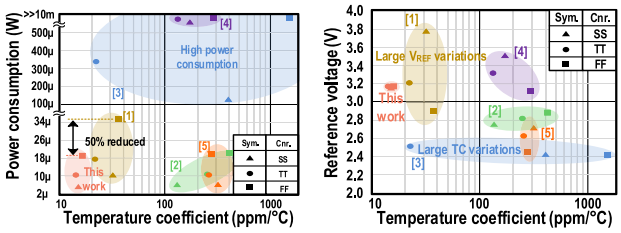


(a) (b)

Fig. 2. (a) Comparisons of power consumption and TC with state of the arts. (b) Comparisons of V_{REF} and TC with state of the arts.

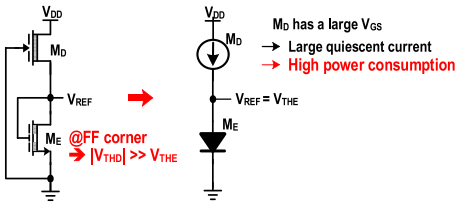


Fig. 3. Conventional VRG in [3] and its problems.

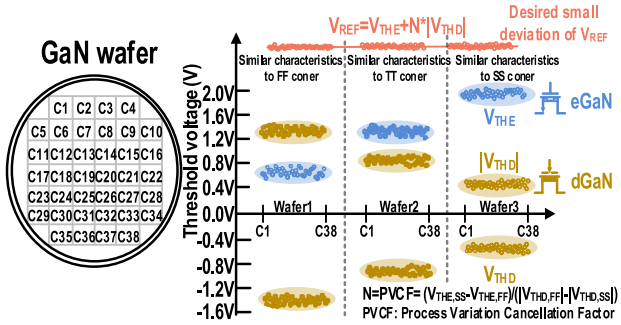


Fig. 4. Threshold voltage of eGaN and dGaN in different process corners.

some prior arts, and N is used to derive the differentiation of V_{REF} with respect to temperature equals zero

$$\frac{\sigma V_{\mathrm{REF}}}{\sigma T} = 0. \tag{2}$$

Since the structural difference between eGaN and dGaN is only the pGaN layer, both V_{THE} and V_{THD} increase from the FF corner to the SS corner. However, as shown in Fig. 4, V_{THD} is always negative and the trend of $|V_{\mathrm{THD}}|$ is opposite compared to V_{THE}. To minimize the deviation in V_{REF} under process variation, N (the scaling factor for $|V_{\mathrm{THD}}|$) should be equal to the process variation cancelation factor (PVCF) as defined in

$$\mathrm{PVCF} = \frac{V_{\mathrm{THE,SS}} - V_{\mathrm{THE,FF}}}{|V_{\mathrm{THD,FF}}| - |V_{\mathrm{THD,SS}}|}. \tag{3}$$

However, in previous works, N is usually different from PVCF to achieve low TC. For example, Fig. 5 shows the architecture of [1], and its V_{REF} can be expressed as

$$V_{\mathrm{REF}} = V_{TH,E1} + \frac{R_2 + R_3}{R_1} \times |V_{TH,D1}| \tag{4}$$

where M equals one and N equals $(R_2+R_3)/R_1$.

For low TC, the value of $(R_2 + R_3)/R_1$ can be derived from (5) and (6), where α and β are the positive TC of V_{THE} and the negative TC of $|V_{\mathrm{THD}}|$, respectively

$$\frac{\sigma V_{\mathrm{REF}}}{\sigma T} = \alpha + \beta \times \frac{R_2 + R_3}{R_1} = 0 \tag{5}$$

$$\frac{R_2 + R_3}{R_1} = -\frac{\alpha}{\beta}. \tag{6}$$

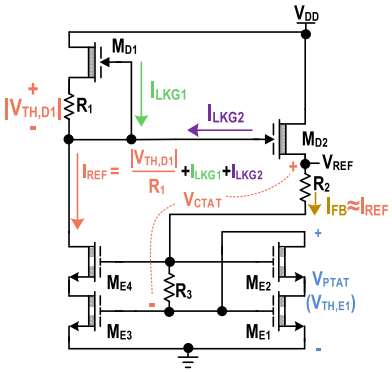
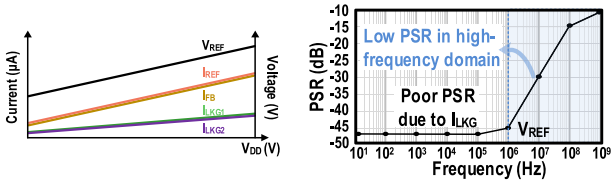


Fig. 5. Conventional VRG in [1].



(a) (b)

Fig. 6. (a) I_{LKG} affects line regulation in [1]. (b) I_{LKG} affects PSR in [1].

For situations where the V_{REF} deviation at different process corner points is small, the value of $(R_2 + R_3)/R_1$ should also meet the requirements of (7). Since it is contradictory for $(R_2 + R_3)/R_1$ to satisfy (6) and (7) at the same time when low TC is ensured first, the V_{REF} range of [1] is 2.88–3.79 V due to process changes. In contrast, Bau et al. [2] and Bimbi et al. [5] first set N equal to PVCF to

$$\frac{R_2 + R_3}{R_1} = \mathrm{PVCF} \tag{7}$$

mitigate process variations. Since N cannot equal $-\alpha/\beta$, TC has a wider distribution.

Furthermore, none of the VRGs in [1], [2], [3], [4], and [5] considered the gate leakage current of dGaN (I_{LKG}), which is much larger than that of the silicon process, causing a huge impact on line regulation and TC. In [1], as V_{DD} increases, I_{LKG1} and I_{LKG2} of M_{D1} and M_{D2} rise, thereby increasing I_{REF}, as shown in Fig. 6(a). A mirrored I_{FB} with the same incremental trend will also increase V_{CTAT}, thus consuming a lot of power and increasing TC with higher V_{DD}. As a result, the PSR and line regulation of V_{REF} degraded, showing that the VRG cannot operate over the wide V_{DD} range in EV power systems. The poor PSR shown in Fig. 6(b) indicates that the previous VRG is not suitable for high-switching switch-mode power supplies (SMPSs) in EV power systems because high-switching coupling noise in V_{DD} affects the reference voltage.

III. Circuit Implementation

Fig. 7 shows the proposed low-TC VRG that can suppress PVT variations [9]. To minimize power consumption, M_{D1} and M_{D2} operate in the subthreshold region to obtain the $|V_{\mathrm{sub}}|$, the source-to-gate voltage (V_{SG}) at the subthreshold region. From Fig. 8, the source current of dGaN (I_S) in the subthreshold region (I_{SL}) increases with temperature, so $|V_{\mathrm{sub}}|$ exhibits a PTAT characteristic, which differs from the CTAT characteristic of $|V_{\mathrm{THD}}|$ (V_{SG} at saturation region). Fig. 9(a) shows that V_{THE} and $|V_{\mathrm{sub}}|$ are PTAT with TC of α and β_{sub}, respectively. For the requirement of CTAT voltage, $R_{2\mathrm{DEG}}$ (composed of a high-density 2-D electron gas (2DEG) layer with a strong PTAT characteristic) is used to generate a current I_{CTAT}. Since

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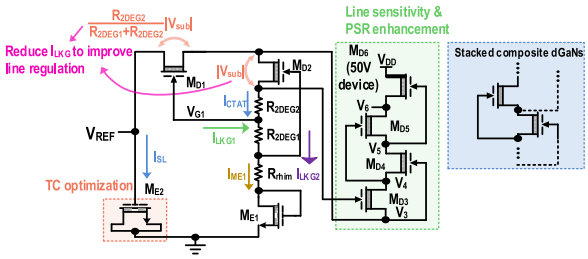


Fig. 7. Proposed GaN-based VRG.

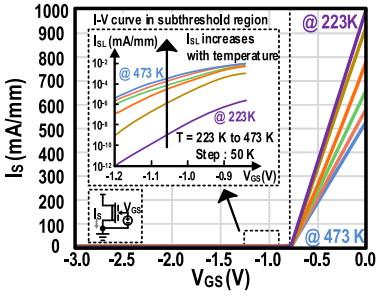
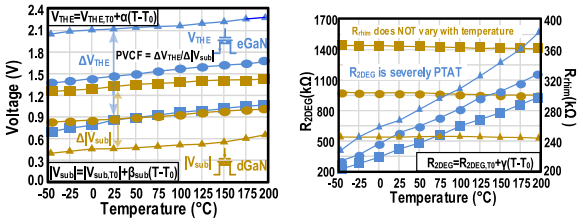


Fig. 8. I–V curve of dGaN at different temperatures.



(a) (b)

Fig. 9. (a) Threshold voltage of eGaN and dGaN at different temperatures. (b) Resistance of R_{2DEG} and R_{rhim} at different temperatures.

the TC γ of R_{2DEG} in Fig. 9(b) is larger than β_{sub}, I_{CTAT} exhibits CTAT characteristics

$$I_{CTAT} = \frac{|V_{sub}|}{R_{2DEG1} + R_{2DEG2}}. \tag{8}$$

A CTAT voltage V_{CTAT} will be across R_{rhim} (composed of a silicon-chromium-based thin-film resistor with nearly zero TC of resistance)

$$V_{CTAT} = R_{rhim} \times \frac{|V_{sub}|}{R_{2DEG1} + R_{2DEG2}}. \tag{9}$$

Thus, V_{REF} of the proposed circuits is derived in

$$V_{REF} = V_{TH,E1} + m \times |V_{sub}| + n \times |V_{sub}| \tag{10}$$

where

$$m = 1 + \frac{R_{2DEG1}}{R_{2DEG1} + R_{2DEG2}} \text{ and } n = \frac{R_{rhim}}{R_{2DEG1} + R_{2DEG2}}.$$

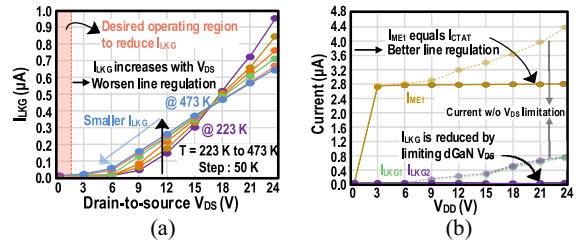
Factor m contributes an additional PTAT design parameter to ensure $N(= m + n)$ meets the requirements for low TC and PVCF. The proper m and n can be derived by the system of linear equations in (11) and (12) with two unknown parameters m and n

$$0 = \frac{\sigma V_{REF}}{\sigma T} = \alpha + m \times \beta_{sub} + n \times \frac{-(\gamma - \beta_{sub})|V_{sub,T_0}|}{(T\gamma - T_0\gamma + 1)^2} \tag{11}$$

$$m + n = \text{PVCF}. \tag{12}$$

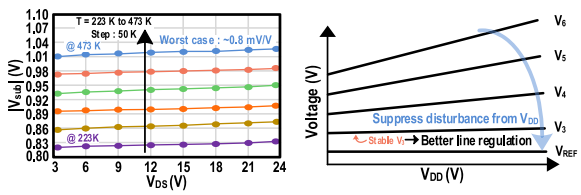
Therefore, a low TC V_{REF} free from process variations can be guaranteed.

According to the relationship between I_{LKG} and the drain-to-source voltage (V_{DS}) of dGaN in Fig. 10(a), to improve PSR, the V_{DS}



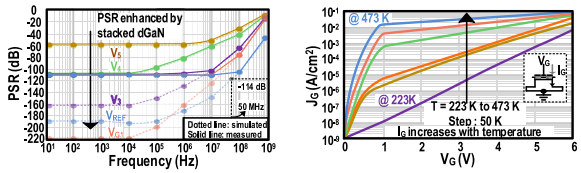
(a) (b)

Fig. 10. (a) Relationship between I_{LKG} and V_{DS} of dGaN at different temperatures. (b) Currents with and without limiting the V_{DS} of dGaN.



(a) (b)

Fig. 11. (a) Relationship between $|V_{sub}|$ and V_{DS} of dGaN at different temperatures. (b) Voltage trend of each node when V_{DD} increases.



(a) (b)

Fig. 12. (a) Each node's simulated and measured PSR. (b) Relationship between gate current density and gate voltage of eGaN.

of dGaN must be less than 1.5 V to achieve a lower variation of I_{LKG}. Therefore, M_{D3} is stacked on M_{D2} to generate a new supply voltage V_3, limiting the V_{DS} of M_{D1} and M_{D2} under a $|V_{sub}|$ (<1.5 V). Thus, I_{LKG}s can be effectively reduced when V_{DD} increases, significantly enhancing line regulation and power consumption, as shown in Fig. 10(b). Fig. 11(a) shows the relationship between $|V_{sub}|$ and the V_{DS} of dGaN. Since $|V_{sub}|$ has only a small increment when V_{DS} increases, dGaN has a great ability to immune disturbances on V_{DD}. To further reduce the influence of V_{DD} variations, multiple stacked composite dGaNs are used to obtain improved PSR and line regulation. After stacking four composite dGaNs, the disturbance from V_{DD} is suppressed stage by stage. V_3 and V_{REF} can be nearly free from V_{DD} variations, as shown in Fig. 11(b). The PSR is also greatly improved by the multiple-stacked composite dGaN and the elimination of I_{LKG}, as shown in Fig. 12(a). The great PSR performance allows the proposed circuits to be used in high-switching EV power systems.

Fig. 8 illustrates the PTAT characteristics of the undesired leakage current (I_{SL}) in the subthreshold region when the dGaN device is nearly turned off, which limits how low the TC of V_{REF} can be achieved. The PTAT behavior of the eGaN gate current (I_G), shown in Fig. 12(b), counterbalances the PTAT I_{SL} to stabilize the operating point, thereby optimizing the TC of V_{REF} while maintaining low-power consumption.

IV. Experimental Results

The proposed GaN VRG is fabricated in 0.5-μm GaN process with 600 × 240-μm^2, as shown in Fig. 13(a). With $m = 1.3$ and $n = 0.35$, the untrimmed V_{REF} and its distribution from 100 sample chips, operated in 6 V V_{DD} at room temperature are shown in Fig. 13(b). The average V_{REF} is about 3.158 V, and its standard

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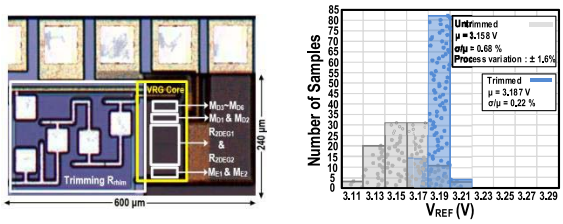
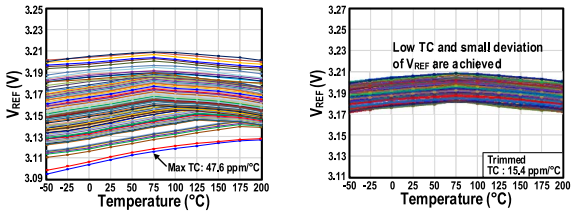


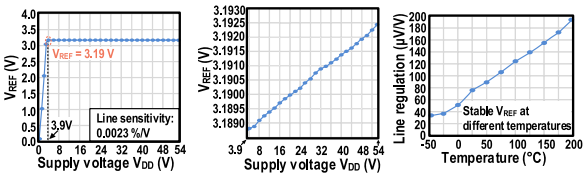
(a) (b)

Fig. 13. (a) Chip micrograph. (b) Measured V_{REF} in 100 samples.



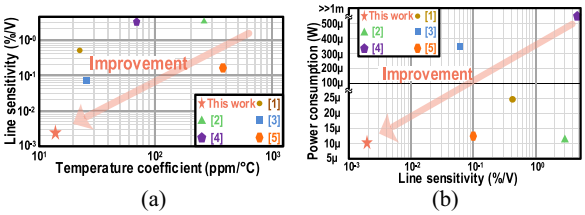
(a) (b)

Fig. 14. (a) Untrimmed V_{REF} from −50 °C to 200 °C of 100 sample chips. (b) Measured V_{REF} from −50 °C to 200 °C of 100 sample chips.



(a) (b) (c)

Fig. 15. (a) Measured V_{REF} in different V_{DD}. (b) Zoom-in of 3.9 V and above voltage. (c) Line regulation versus temperature.



(a) (b)

Fig. 16. (a) Comparisons of line sensitivity and TC with state of the arts. (b) Comparisons of power consumption and line sensitivity with state of the arts.

deviation is 0.68%. Furthermore, the process variation is only 1.6% without trimming. Laser trim is adopted to fine-tune the resistance of R_{rhim}, and the 2-point trimming method is used to optimize TC and offset. After trimming R_{rhim}, the average V_{REF} is about 3.187 V, and its standard deviation is further reduced to 0.22%. Fig. 14 shows the untrimmed and trimmed V_{REF} measured for 100 samples over the range −50 °C to 200 °C, where the maximum measured TC of untrimmed V_{REF} and trimmed V_{REF} are 47.6 and 15.4 ppm/°C, respectively. Ceramic packaging and ceramic board that can withstand over 300 °C are adopted in measurement to operate at high temperatures. Fig. 12(a) shows the simulated and measured PSR, which is stable in the low-frequency domain. After PSR enhancement, the simulated PSR reaches −187 dB at 100 Hz and −114 dB at 50 MHz, exhibiting the highest PSR at tens of megahertz compared to prior arts. Due to the accuracy of the measuring instruments, the measured PSR of some nodes is significantly different from the simulated results. Fig. 15(a) shows that at a minimum V_{DD} of 3.9 V, V_{REF} is 3.19 V. Line sensitivity for V_{REF} is 0.0023%/V at room temperature over a V_{DD} range of 3.9 to 48 V. Fig. 15(c) shows the line regulation at different temperatures, which is less than 200 μV/V. Fig. 16(a) and (b) compare TC, line sensitivity, and power consumption with state of the arts. With low power consumption, the line sensitivity is reduced by 96.3% compared to [3], and TC is reduced by 34.7% compared to [1]. Table I shows the overall performance and compares this letter with conventional GaN-based VRGs. The proposed circuit is the only one that can solve the problems caused by process variations and achieve low TC at the same time.

TABLE I
COMPARISON WITH STATE OF THE ARTS

Design	This work	ISSCC'20 [1]	APEC'23 [2]	EDL'22 [3]	ISPSD'09 [4]	Electronics'22 [5]
Technology (μm)	0.5	0.5	0.5	N/A	N/A	0.5
Process	GaN HEMT	GaN HEMT	GaN HEMT	GaN MIS-HEMT	GaN HEMT	GaN HEMT
Power (μW)	10.9	24.96*	11.50	330	1000	10.53
V_{DD} Range (V)	3.9~54	3.9~24	3.1~20	4.8~50	4~10	3.9~24
V_{REF} (V)	3.19	3.19	2.81	2.53	2~8	2.7
Line Sensitivity (%/V)	0.0023	0.32	2.2	0.063	N/A	0.105
PSR @ 100Hz (dB)	<187 (-114 @ 50MHz)	-45	N/A	-42.8	0	-45
Temp. Range (°C)	-50~200	-50~200	25~150	-25~250	25~250	-40~190
TC (ppm/°C)	15.4	23.6	256	26.2	70, 135**	>200
Process Variation Cancellation	Yes	No	Yes	No	No	Yes
Process Variation (%)	1.6%	N/A	4.4%	N/A	N/A	N/A
Die Size (mm²)	0.144 (0.033 (Core))	0.52	0.006	0.104	0.315	0.05

*after correction **simulation results in this work

V. CONCLUSION

The proposed VRG is the first GaN VRG to exhibit both small V_{REF} deviation and low TC, only 15.4 ppm/°C from −50 °C to 200 °C range. By eliminating I_{LKG} and utilizing multiple stacks of composite dGaN, line sensitivity and PSR are greatly improved. The low line sensitivity (only 0.0023%/V from 3.9 to 48 V) and high PSR enable the proposed VRG to operate properly in high-voltage and high-switching EV power systems. All performance improvements are achieved with a low power consumption of 10.9 μW.

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